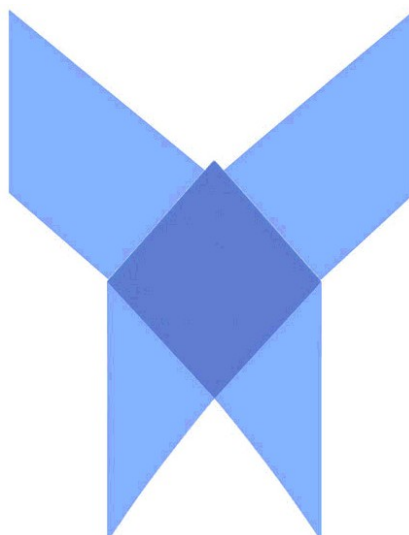




Islamic Azad University
Tehran

Faculty of Computer Engineering
Course Description
Computer Engineering
Bachelor's Degree



Remarks:

According to the Academic Instruction of University, in the academic system, each theoretical credit is presented in 16 or 17 hours, each practical credit in 32 or 34 hours, each of the workshop credit in 48 or 51 (except Professional Certificates) and each internship or training credit in 64 or 68 hours.

There are some different type of the courses passing by a student at this university, as listed below:

1. Basic
2. Core / Major
3. Specialized
4. Elective / Optional
5. General
6. Internship / Workshop
7. Project / Thesis

Each one of these types contain some courses to be passed, according to the educational regulations passed by the Ministry of Science, Research and Technology or Ministry of Health & Medical Education of Iran.

The grading system in this university is from 0 to 20. The minimum passing grade for a course leading to an Associate's Degree or a Bachelor's Degree is 10, for a course leading to a Master's Degree and Medical Sciences is 12 and for a course leading to a PhD Degree is 14.

Note: This document has been prepared at **Pouyan Fallahi's** request in the **Course Description** Platform. Scan the QR-code below to certify this document.



www.coursedescription.ir



Title of the Course: Persian language

Number of Credits: 3

Type of the Course: General | Theoretical

Training hours: 48h Theoretical

Course Objectives:

Familiarity with Persian language and reading literary texts, poetry and prose in Iranian Persian literature.

Title of the Course: Foreign Language For Specific Purpose

Number of Credits: 3

Type of the Course: General | Theoretical

Training hours: 48h Theoretical

Course Objectives:

The content of this course is presented based on the acquisition of skills in reading and understanding the concepts of foreign texts in order to prepare students for the placement of semi-specialized texts.

Syllabus:

A review of the structures and vocabulary of reading different texts. Emphasis on learning skills including: A- How to use English dictionary and encyclopedia. B - Vocabulary. C - Writing outlines. D - Summarizing. E- Fast reading different texts.

I.A.U



Title of the Course: Islamic Ideology I (origin & Resurrection)

Number of Credits: 2

Type of the Course: General | Theoretical

Training hours: 32h Theoretical

Course Objectives:

Explaining the need for Muslim students to deal with religion and religious categories, deepening and expanding information and strengthening the belief in God and faith of students in the field of theology and resurrection.

Syllabus:

Man and faith: spiritual needs of man, the role of religious faith in human life, reason and faith. The concept of God: different perceptions about God, God in Abrahamic religions (Judaism, Christianity, Islam). Finding God: heart and intuition, intellect and reason. Monotheism and polytheism: monotheism of nature and attributes, monotheism of verbs, monotheism of worship (invocation and intercession). Other attributes of God: divine justice, wisdom and judgment, the problem of evil. Predestination and discretion (Qada and Qadr). Resurrection and immortality of man: death and purgatory, resurrection.

Title of the Course: Discrete Mathematics

Number of Credits: 3

Type of the Course: Core | Theoretical

Training hours: 48h Theoretical

Course Objectives:

This course covers elementary discrete mathematics. It emphasizes mathematical definitions and proofs as well as applicable methods. Topics include formal logic notation, proof methods; induction, well-ordering; sets, relations; elementary graph theory; integer congruences; asymptotic notation and growth of functions; permutations and combinations, and counting principles.

TextBooks:

- 1- R. P. Grimaldi, Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, Pearson, 2017.
- 2- K. H. Rosen, Discrete Mathematics and Its Applications, 7th Edition, McGraw-Hill, 2011.
- 2- S. S. Epp, Discrete Mathematics with Applications, 4th Edition, Brooks Cole, 2010.



Title of the Course: Islamic Ideology II(Prophet Hood & Imamate)

Number of Credits: 2

Type of the Course: General | Theoretical

Training hours: 32h Theoretical

Course Objectives:

Expanding the awareness of students in the field of religion, mission, Islam, Imamate and Wilayat.

Syllabus:

The background of religion and prophethood: the definition of religion and its background in history, Judaism and Christianity, their origin and destiny. Getting to know the Torah and the Bible and comparing them with the Qur'an. The influence of Christianity in Western society and its comparison with the influence of Islam in the birth of Islamic civilization. Goals, dimensions and scope of religion: the necessity of revelation and prophethood for spiritual happiness and worldly life, the infallibility of prophets, the role of religion in worldly life, the common gem of religion and the secret of the plurality of religions and laws, the relationship between science and religion. Knowledge of Islam: Qur'an and Sunnah, miracle of Qur'an, validity of Sunnah, authority and similarities, reason and its place in knowledge of religion, completeness and response of Islam to changing human needs, method of understanding religion. Imamate and Wilayat: The meaning of Imamate and Wilayat, the dimensions and affairs of Imamate (religious reference, political Wilayat, spiritual Wilayat), the infallibility of Imams and the reasons for their installation, Mahdism. Authority and Wilayat in the Age of Absence: Religious Authority in the Age of Absence, Wilayat Faqih and Political Leadership in the Age of Absence. History of Velayat al-Faqih: the basis of the legitimacy of the religious government and the authority of the legal guardian, installing, dismissing and supervising the legal guardian.



Title of the Course: Physics 1

Number of Credits: 3

Type of the Course: Basic | Theoretical

Training hours: 48h Theoretical

Course Objectives:

The aim of this course is to introduce the concepts of classical mechanics and thermodynamics.

Syllabus:

Measurement: science and measurement, unit devices, SI units, mass, length, time, dimensional analysis. Vector analysis: addition and subtraction of vectors, vector multiplication (internal and external multiplication). Motion kinematics: definitions of displacement, velocity and acceleration, one-dimensional motion (uniform motion and motion with constant acceleration), two-dimensional motion (throwing motion and circular motion). , relative speed and relative accelerations (Galilean relativity). Dynamics of motion: Newton's laws including the first law, second and third laws (concepts of mass and force). Laws of force: weight force, Hooke's elastic force, friction force, string tension, etc. Work and energy: work of constant force, work of variable force, principle of work (work-energy theorem), power, work of internal forces. Consistency of energy: stable and unstable forces, constancy of mechanical energy, constancy of energy. Impact and impact: impact, elastic collisions and one-dimensional inelastic, two- and three-dimensional collisions. Kinematics of rotational motion: displacement, angular velocity, angular acceleration, angular quantities as vectors, force torque, angular momentum, energy of rotational motion. Dynamics of rotational motion: equations of rotational motion, rotation about an axis constant, planar rolling, law of conservation of angular momentum, thermometry: thermal equilibrium, zeroth principle of thermodynamics, thermometric parameter, thermometric equation, thermometric standard points, complete gas temperature scale, comparison of Fahrenheit and Celsius, thermal expansion. Heat: thermal energy, ways of heat transfer, heat and work, the first law of thermodynamics, simple applications of the first law, equation of state of a perfect gas (macroscopic description), different processes on a perfect gas (isothermal, constant volume, constant pressure, direct). Kinetic theory of gases: Equation of state of perfect gas (microscopic description), calculation of pressure, change of temperature based on kinetic energy of molecules, specific heat of perfect gas, degrees of freedom and equal distribution of energy, Maxwell's distribution function for molecular speeds. Second law of thermodynamics: reversible and irreversible processes, cyclic Carnot, second law of thermodynamics, efficiency of machines, refrigerators, entropy, entropy and second law, statistical view of entropy.

TextBooks:

- 1- D. Halliday, R. Resnick, and J. Walker, Fundamentals of physics, John Wiley & Sons, 2013.
- 2- R.A. Serway, and C. Vuille, College physics, Cengage Learning, 2014.
- 3- H. D. Young, R.A. Freedman, and R. Bhathal, University physics, Pearson Higher Education AU, 2010.



Title of the Course: General Mathematics 1

Number of Credits: 3

Type of the Course: Basic | Theoretical

Training hours: 48h Theoretical

Course Objectives:

Basic concepts of Calculus and Geometry will be taught to the students in this course which provides necessary background for technical courses.

Syllabus:

Cartesian coordinates; polar coordinates; complex numbers; addition, product, root & geometrical representation of complex numbers; polar representation of complex numbers; function; functions algebra; limit and relevant theorems; infinite limit and limit in infinite; left-hand and right-hand limit; connectivity; derivative; derivation formula; inverse function and its derivative; trigonometric functions derivative and their inverse functions; Rolle's theorem; mean theorem; Taylor expansion; geometrical and physical applications of derivative; curves and acceleration in polar coordinates; application of derivative in approximation of equations roots; definition of integral of continuous functions and piecewise continuous; basic theorems of differential & integral arithmetic; primitive function; approximate methods of integral estimate; application of integral in computation of area, length of curve, moment, center of gravity and labor(in Cartesian and polar coordinates); logarithm and exponential function and their derivative; hyperbolic functions; integration methods such as change of variable, component and decomposition of fractions; transform of special variables of sequence and numerical series and relevant theorems; power series and Taylor theorem with remainder.

TextBooks:

R.L. Finney, G.B. Thomas, Calculus and Analytic Geometry, Geometry, 9th Edition, Addison Wesley, 1996

I.A.U



Title of the Course: Basics of Computer and Programming

Number of Credits: 3

Type of the Course: Core | Theoretical

Training hours: 48h Theoretical

Course Objectives:

This course will teach students the main concepts of Programming using a Programming Language such as C++. Techniques of Developing and Implementing Algorithms in a programming language will be taught in this course.

Syllabus:

Introducing organization and main parts of computer, machine language and assembly, numeric and non-numeric data representation, algorithms and sub-algorithms and flowchart, basic concepts such as frequency, selection, repetition and branching, familiarity with a structured language and programming including: constants and variables, computational and logical phrases, different types of instructions, different types of conditional operations loops, vectors and matrices, subprograms, input and output instructions, common algorithms such as methods of search and arrangement, practical examples of programming in C++ Programming.

TextBooks:

1. T.C. Bartee, Digital Computer Fundamentals, McGraw Hill, 1981
2. J.G. Brookshear, Computer Science, and Overview, 6th Edition, Addison Wesley, 1999

I.A.U



Title of the Course: Specialized Language(Computer)

Number of Credits: 2

Type of the Course: Core | Theoretical

Training hours: 32h Theoretical

Course Objectives:

To enhance students' reading comprehension of technical computer science texts. To develop vocabulary of specialized computer science terms. To improve skills in translating technical documents from English to Persian. To enable students to write abstracts and summaries in English. To prepare students for understanding academic papers and manuals in software engineering.

Syllabus:

Introduction to technical English and computer science terminology. Reading and analyzing texts on computer architecture. Vocabulary related to operating systems. Comprehension of networking and internet protocols. Specialized terms in database management systems. Software engineering concepts and terminology. Reading algorithm descriptions and data structures. Translation techniques for technical documents. Writing technical abstracts and summaries. Understanding security and cryptography terms. Artificial intelligence and machine learning vocabulary. Cloud computing and distributed systems terminology. Human-computer interaction and user interface terms. Software testing and quality assurance vocabulary. Project management and agile methodology terms. Review of advanced topics and exam preparation.

I.A.U



Title of the Course: Differential Equations

Number of Credits: 3

Type of the Course: Basic | Theoretical

Training hours: 48h Theoretical

Course Objectives:

In this course, first and second levels of linear differential equations and some nonlinear differential equations will be introduced, in addition, students will learn about some numerical and analytical ways to solve Mathematical Problems.

Syllabus:

Nature of differential equations and their solution, family of graphs and vertical routes, physical patterns, separable equation, first order linear differential equation, homogeneous equation, second order linear equation, homogeneous equation with fixed constants, method of indefinite constants, method of changing parameters, application of second order equations in physics and mechanics, solution of differential equation with series, Bessel and Gamma functions, Legendre polynomial, an introduction to differential equations set, Laplace transform and its application in solving differential equations.

TextBooks:

1. E. Kreyszig, Advanced Engineering Mathematics, 1999
2. P. Blanchard, R.L. Devaney, and Co. Hall, Differential Equations, 1st Edition, Brooks/ColePub, 1998

I.A.U



Title of the Course: General Mathematics 2

Number of Credits: 3

Type of the Course: Basic | Theoretical

Training hours: 48h Theoretical

Course Objectives:

After completing the course, the student will find the appropriate mental background to understand and solve various engineering problems and acquire the ability to calculate the vectors, determinants, matrices, partial derivatives, complete differentials, and spherical and cylindrical coordinates.

Syllabus:

Parametric equations; space coordinates; vector and space; numerical product; matrix 3×3 of three-indeterminate linear equations system; operation on lines; matrix reverse; solving equations system; linear independence; base in R^2 ; R^3 linear transform and its matrix; determinate 3×3 and characteristic value and vector; vector product; second order line and plane equations; two vector functions and its derivative; speed and acceleration; bending; normal vector to a curve; multivariable function; directional and partial derivative; tangent plane and normal line to a curve; multivariable function; directional and partial derivative; tangent plane and projecting line of gradient; chain of rule for partial derivative; exact differential; second kind and third kind integrals and their application in geometrical and physical problems; transform of integration arrangement (without accurate affirmation); cylindrical and spherical coordinates; vector field; curvilinear integral; surface integral; divergence; curl; Laplacian; potential of green space and divergence and stochastic.

TextBooks:

R.L. Finney, G.B. Thomas, Calculus and Analytic Geometry, Geometry, 9th Edition, Addison Wesley, 1996

I.A.U



Title of the Course: Islamic Ethics (Basics and Concepts)

Number of Credits: 2

Type of the Course: General | Theoretical

Training hours: 32h Theoretical

Course Objectives:

Familiarizing students with the basics and concepts of Islamic ethics in the field of moral virtues and vices in order to acquire virtues and avoid moral vices.

Syllabus:

General. The concept, importance and position of Islamic ethics. History of Islamic ethics. Different approaches in Islamic ethics. Basics of Islamic ethics. Purposefulness of human life. Proximity factors and obstacles. Degrees of closeness to God. Self Knowledge. Dimensions of human existence (sublime and material dimension) and his immortality. Freedom and choice. The role of belief and action in the development of human personality. Moral virtues and vices: in the relationship between man and God (spirituality and servitude), in relation to himself (individual morality), in relation to others (social morality). Moral virtues: love for God and his saints, trust, contentment, remembrance of death, contemplation, sincerity, fear and hope. Moral vices: worldliness, following the whims of the ego, etc.

Title of the Course: Introduction to Holly Defense

Number of Credits: 2

Type of the Course: General | Theoretical

Training hours: 32h Theoretical

Course Objectives:

The history of the Iran-Iraq war.

I.A.U



Title of the Course: Physics 2 -Lab

Number of Credits: 1

Type of the Course: Basic | Practical

Training hours: 32h Practical

Course Objectives:

The purpose of this course is to learn electrical and electronic measuring devices and to perform basic tests of electrical circuits.

Syllabus:

Identification of an oscilloscope. Identification of a galvanometer and converting it into an ammeter, voltmeter, and wattmeter. Plotting the characteristic curve of bipolar and tripolar lamps. Plotting the characteristic curve of diodes and transistors. Measuring the capacitance of capacitors.

TextBooks:

1. D. Halliday and R. Resnick, Fundamentals of Physics, 10th Edition, John Wiley Sons, 2014.
2. R. A. Serway, and C. Vuille, College Physics, 10th Edition, Cengage Learning, 2015.
3. H. D. Young, R. A. Freedman and L. Ford, University Physics, 14th Edition, Pearson Education Limited, 2016.

Title of the Course: Family and Population Knowledge

Number of Credits: 2

Type of the Course: General | Theoretical

Training hours: 32h Theoretical

Course Objectives:

Familiarity of students with the basics of family knowledge and how to form, consolidate, and excel it and demographic characteristics and their role in improving the quality of life

I.A.U



Title of the Course: Data Structure

Number of Credits: 3

Type of the Course: Core | Theoretical

Training hours: 48h Theoretical

Course Objectives:

Familiarity with Information Structures, Effects of Structures in Produced Applications, Selecting Optimized Inside-Memory Structures, Organizing Memory based on requirements.

Syllabus:

Arrays, vectors, matrices, private matrices, arrays display, stacks, queues and rows, bond lists: graphic, cyclical, double bond, multi-bond, method of display and application of bond lists, trees and their menstruation, method of representation and application of trees: decision making trees, search trees, tree of the game and etc., graphs and their representation, dynamic memory allocation and the relevant issues, searching and sorting and combining algorithms

TextBooks:

1. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, Introduction to Algorithms, Third Edition, The MIT Press, Cambridge, Massachusetts 2009
2. A.V. Aho, J.E. Hopcroft, J.D. Ullman, Data Structures and Algorithms, Addison Wesley, 1983
3. D.E. Knuth, The Art of Computer Programming, Volume I: Fundamental Algorithms, 3rd Edition, Addison Wesley, 1997

I.A.U



Title of the Course: Algorithms Design

Number of Credits: 3

Type of the Course: Core | Theoretical

Training hours: 48h Theoretical

Course Objectives:

The main purpose of this course is to teach the basic concepts and common methods of analysis and design of algorithms. In this lesson, students learn how to analyze a given problem and find some possible algorithms to solve it. Then analyze and compare those algorithms in terms of computational complexity and identify the best ones based on the characteristics of the input samples. In this lesson, basic algorithms for solving some common problems will be introduced.

Syllabus:

Review of essential points of Data Structures, problem solving methods (for each methods some problems and special algorithm for the problem shall be represented and being analyzed), introducing to complexity, divide & conquer method (problems to be analyzed: max and min of an array, multiplication of two n-bit number, Strassen method about matrix multiplication, round robin algorithm, sorting with Quicksort algorithm), dynamic programming method (matrix multiplication, traveling sales man problem, polygon triangulation), greedy algorithms(scheduling problem, Huffman Code, making change), methods based on exhaustive search, alpha-beta pruning (puzzle, tic-tac-tac), revelation of methods for problems, graph algorithms(searching methods of graph, Dijkstra algorithms, minimum spanning tree, Floyd algorithm, topological sorting and...), maximum flow networks and other problems.

TextBooks:

- 1- Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, Introduction to Algorithms, Third Edition, The MIT Press, Cambridge, Massachusetts 2009.
- 2- R.E. Neapolitan and K.Naimipour, Foundations of Algorithms using C++ Pseudo Codes, 2nd Edition, Johns and Bartlett Publishers, 1998.



Title of the Course: Logic Circuits

Number of Credits: 3

Type of the Course: Core | Theoretical

Training hours: 48h Theoretical

Course Objectives:

The purpose of this course is to get acquainted with the principles and components of logic circuits and how these circuits work, to acquire skills in designing circuits and digital systems and skills in analyzing digital circuits in terms of their operation and speed of operation.

Syllabus:

Logic level models, including Boolean algebra, finite state machines, arithmetic circuits, and hardware description languages. Logic gates, memory, including CMOS gates, flip-flops, arrays, and programmable logic. Both combinational and sequential logic circuits are covered in this course. The emphasis is on the use of Boolean algebra and basic logic gates to build cost effective complex logic circuits.

TextBooks:

Victor P. Nelson, H. Troy Nagle, Bill D. Carroll and David Irwin, Digital Logic Circuit Analysis and Design, Prentice Hall Inc., 1996

Title of the Course: Theory of Languages and Machines

Number of Credits: 3

Type of the Course: Core | Theoretical

Training hours: 48h Theoretical

Course Objectives:

The course introduces some fundamental concepts in automata theory and formal languages including grammar, finite automaton, regular expression, formal language, pushdown automaton, and Turing machine.

TextBooks:

- 1-P. Linz, Introduction to Formal Languages and Automata, Second Edition, D. C. Heath Company, 1996
- 2-D. Wood, Theory of Computation, Prentice-Hall, 1986
- 3-P. Reveseze, Theory of Formal Languages, MC Graw-Hill, 1985
- 4-D. I. A. Cohen, Introduction to Computer Theory, John Wiley & Sons, Inc., 1991



Title of the Course: History of culture and civilization of Islam and Iran

Number of Credits: 2

Type of the Course: General | Theoretical

Training hours: 32h Theoretical

Course Objectives:

Familiarity with the formation of Islamic civilization and internal and external elements effective in its advancement and decline in order to strengthen self-confidence and consolidate Islamic national identity.

Syllabus:

Basic topics. Terminology of cultural history, civilization and modernity. The constituent elements of culture and civilization. Definition of Islamic civilization and its historical and geographical scope. Islamic civilization and its causes and factors. Characteristics of Islamic civilization. Scientific flourishing movement in Islamic civilization. Political, social and scientific institutions of Islamic civilization. Religious, cultural and social causes and factors of the emergence and prosperity of Islamic civilization. The influence of previous cultures and civilizations in the emergence of Islamic civilization (Greece, Iran, etc.). Mutual services of Islam and Iran. Backgrounds of decline, causes and factors of stagnation of Islamic civilization. Invasion of foreign enemies. Aristocracy and autocracy and the caliphate's departure from authentic Islamic standards. Petrification and political and social restrictions. Worldliness and moral degeneration and deviation from true Islam. The influence of Islamic civilization on Western civilization and the emergence of the Renaissance. Available capacities in the Islamic world. Geographical and geopolitical position of Islamic countries. Underground and human resources of Islamic countries. Cultural and spiritual capital of Islam. Spiritual degeneration and internal crises of the modern world.

I.A.U



Title of the Course: Physics II

Number of Credits: 3

Type of the Course: Basic | Theoretical

Training hours: 48h Theoretical

Course Objectives:

The purpose of this course is to familiarize with the basics of electricity and magnetism and to use these basics to solve elementary physics problems.

Syllabus:

Electricity. Coulomb's law: electric charge, conductors, semiconductors and non-conductors, conservation of electric charge, Coulomb's law (vector formulation), applications of Coulomb's law. Electric field: concept of field, calculation of field resulting from discrete charge distribution, concept of continuous charge distribution, field resulting from distribution Continuous charge, charge movement in an electric field, electric dipole in an electric field. Gauss's law: Faraday's experiment and the concept of electric flux, Gauss's law and its applications. Electric potential: electric potential difference, electric unipolar and electric dipole potential, potential of a continuous charge distribution, Persistence of electrostatic field, concept of electric potential energy (discrete charges and continuous distribution). Capacitors and insulators: definition of capacitor, calculation of capacitor capacity (flat, cylindrical, spherical), connection of capacitors (parallel, series), energy stored in capacitor, capacitor Content of insulation, polarization of insulations, Gauss' law and insulations, electric current: current density, resistance and Ohm's law, electric power and energy in an electric field, electric circuits: electric driving force, simple circuits (series (series) and parallel resistances), voltage law Kirchhoff, Kirchhoff's current law, analysis of multi-loop circuits, electrical measuring devices (voltmeter, ammeter, ohmmeter, potentiometer), RC circuits. Magnetic field: concept of magnetism, magnetic force, charged particle movement in magnetic field, current loop in magnetic field, applications (cyclotron, Hall effect, electric motors). Electromagnetic induction: induced currents, Faraday's law, magnetic flux, induction and conservation of energy , induced electric field, self-inductance and mutual induction, RL circuits, magnetic energy, oscillation in RL circuit, alternating current circuits: RLC circuits, alternating current sources and phasors, resistance in AC circuit, self-induction in AC circuit, capacitor in AC circuit, series circuit RLC, power in an AC circuit, resonance in a series RLC circuit. Maxwell's equations: basic equations of electricity and magnetism, induced magnetic field, generalization of Ampere's law, Maxwell's equations, electromagnetic waves, wave equation, spectrum of electromagnetic waves. Electromagnetic waves: light as an electromagnetic wave , speed of light, Pointing vector and intensity of light, radiation pressure.

TextBooks:

1. D. Halliday, R. Resnick, and J. Walker, Fundamentals of physics, John Wiley & Sons, 2013.
2. R.A. Serway, and C. Vile, College physics, Cengage Learning, 2014.
3. H.D. Young, R.A. Freedman, and R. Bhathal, University physics, Pearson Higher Education AU, 2010.



Title of the Course: Artificial intelligence and expert systems

Number of Credits: 3

Type of the Course: Specialized | Theoretical

Training hours: 48h Theoretical

Course Objectives:

This course presents a study of artificial intelligence and expert systems. Topics include PROLOG programming, search methods, knowledge acquisition, knowledge representation including belief networks, knowledge validation, neural networks, expert system development including uncertainty management methods such as statistical, symbolic, and fuzzy logic, expert system shell, survey of current expert systems, and future trends.

TextBooks:

1- Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach. 3rd Edition, 2009.

Title of the Course: Microprocessors and assembly Language

Number of Credits: 3

Type of the Course: Core | Theoretical

Training hours: 48h Theoretical

Course Objectives:

In this course, students with a microprocessor architecture (by choosing an architecture such as ARM,) software, hardware and microcontroller components (by choosing a microcontroller from the AVR family, ARM and the like), programming in assembly language and higher levels, How the CPU connects and communicates with memory and peripherals. In this course, students will also learn how to use subprocessors and microcontrollers to build embedded systems, the Internet of Things, and the like.

TextBooks:

1- M. A. Mazidi, S. Naimi, S. Naimi, The AVR Microcontroller and Embedded Systems using Assembly and C, 1st Edition, Prentice Hall, 2010.

2- M. A. Mazidi, D. Causey, R. D. McKinlay, PIC Microcontroller and Embedded Systems using Assembly and C for PIC18, Prentice Hall, 2008.



Title of the Course: Engineering Mathematics

Number of Credits: 3

Type of the Course: Core | Theoretical

Training hours: 48h Theoretical

Course Objectives:

Familiarization with the concepts and applications of series and Fourier transform, fields of occurrence and methods of analysis of equations with partial derivatives, functions and mappings in the complex plane.

Syllabus:

Fourier series and Fourier transform: definition of Fourier series, Euler's formula of half-period expansion, forced functions, Fourier integral, Laplace transform; Gibbs theorem and limits of Fourier analysis; orthogonal and complete functions; series convergence and Fourier transform; application of Fourier analysis in solving engineering problems; symmetric properties of transformation and Fourier series; Introduction of short time Fourier transform. partial equations or derivatives: one-variable wave equation; method of separation of variables; D'Alembert's solution for the wave equation; Heat diffusion equation; wave equation; Laplace's equation in Cartesian, spherical and polar coordinates; Elliptic, parabolic and hyperbolic equations; Application cases of Laplace transform in solving equations with partial derivatives; Solving partial derivative equations using the Fourier integral; Derivation of telegraphic equations. Analytical functions and isometric mapping and complex integrals: limit and continuity; derivative of complex functions; exponential and trigonometric functions; hyperbolic and logarithmic; Inverse and exponential trigonometry with mixed view; maintaining the Dirichlet boundary condition; Conservation of energy under isometric transformation; Application of isometric transformation in electric capacity calculations. Line integral in the complex plane: Cauchy's integral theorem; Calculating the linear integral using indefinite integrals; Cauchy formula; Taylor and McLaren expansions; Calculation of integrals by the method of balances; The remaining case; Calculation of the real integral of the fraction; Calculation of real trigonometric integral.

TextBooks:

- 1- R. Wylic, Advanced Engineering Mathematics. 6th ed., McGraw-Hill, 1995.
- 2- E. Kreyszig, Advanced Engineering Mathematics, 10th ed., Wiley, 2011.



Title of the Course: Electronic circuits

Number of Credits: 3

Type of the Course: Core | Theoretical

Training hours: 48h Theoretical

Course Objectives:

To develop the fundamental tools of linear circuit analysis which will be useful to all engineers. To learn the "alphabet" of circuits, including wires, resistors, capacitors, inductors, voltage and current sources, and operational amplifiers. To prepare students for more advanced courses in circuit analysis.

Syllabus:

Circuit variables and circuit elements. Some circuit simplification techniques. Techniques of circuit analysis. The operational amplifiers. The natural and step response of RL and RC circuits. Natural and step responses of RLC circuits. Sinusoidal steady-state analysis. Introduction to the Laplace Transform. The Laplace Transform in circuit analysis.

Title of the Course: Systems Design and Analysis

Number of Credits: 3

Type of the Course: Specialized | Theoretical

Training hours: 48h Theoretical

Course Objectives:

Just as solving a mathematical problem requires first identifying its complexities and dimensions, designing appropriate strategies to solve the problem and then proceeding to solve it, such as building computer systems for a set of people (which in many cases may be More complex mathematical problems) also need to be analyzed and designed before implementation. In this course, students are introduced to analysis and design skills such as requirements analysis, feasibility analysis, modeling, process analysis, architectural design, and UI / UX. In addition to these skills, this course will include computer systems management and production planning skills such as production methods, process automation, and project management concepts in such projects.

TextBooks:

- 1- L.D. Bentley, K.C. Dittman, and J.L. Whitten. Systems analysis and design methods. Irwin/McGraw Hill, 2007.
- 2- K.S. Rubin. Essential Scrum: A practical guide to the most popular Agile process. Addison-Wesley, 2012.
- 3- M. Fowler, C. Kobryn, and K. Scott. UML distilled: A brief guide to the standard object modeling language. Addison-Wesley Professional, 2004.



Title of the Course: Operating Systems

Number of Credits: 3

Type of the Course: Core | Theoretical

Training hours: 48h Theoretical

Course Objectives:

This course relates to the most of the Operating Systems' fundamental structures, their specification and usages.

Syllabus:

Definition, existence philosophy, basic functions, different types of operating systems, operating systems hierarchy, operating system in view of user, Program Status Word (PSW), concept of interrupts, different types of interrupts and their processing, interrupts priority and next coming, I/O programming and the relevant facilities, concurrency in I/O, memory management, introducing multiprogramming environments, static memory allocation, dynamic memory allocation, commutative memory allocation, paging according to demand, partitioning, paging part, memory hierarchy. Processor management, works scheduling, scheduling policies, processes scheduling, processes scheduling policies, multiprocessor systems, weak communication, stable communication, allocation of sources to processes, competition mode, blocked mode and methods of releasing, mutual exclusion, processes concurrency by use of semaphore and problems resulting from which. Systems management, basic functions, exclusive, common and dummy instruments, I/O instruments and subsidiary memories, conversion of exclusive instruments to dummy instruments, extra linear processing, direct communication, auxiliary processor of spooling system.

TextBooks:

1. W. Stallings, Operating Systems, 4th Edition, Prentice Hall, 2001
2. Silberschatz, J.L. Peterson, Operating Systems Concepts, Addison Wesley, 2000

I.A.U



Title of the Course: Computer Architecture

Number of Credits: 3

Type of the Course: Core | Theoretical

Training hours: 48h Theoretical

Course Objectives:

An introduction to computer configuration, introducing different generations of computer, register transfer and micro-operations, register transfer language, inter-register transfer, computational micro-operations, sliding micro-operations, control functions.

Syllabus:

Basic computer organization and its design, instruction codes, computer instructions, scheduling and controlling, designing a sample computer such as PDP/8 and its micro-operations, methods of numbers representation, representation by fixed point, representation by floating point, other binary codes, errors revealing codes, organization of central processor including systems with several processor registers and bus system and systems using stack, study of several computers such as PDP/11, IBM 370. Designing computational processors, comparison and subtraction of binary numbers without sign, algorithm of multiplication and division with fixed and floating point. -Input and output organization, memory organization, auxiliary memories, memory of microcomputer, hierarchy of memory, associative memory, dummy memory, cache memory, memory management hardware.

TextBooks:

1- Computer System Architecture, M. Morris Mano, Prentice Hall, 1982 2- Computer Architecture, C.C.Foster, Van Nostrand Reinhold Company, 1985

I.A.U



Title of the Course: Logic Circuits and Computer Architecture - Lab

Number of Credits: 1

Type of the Course: Core | Practical

Training hours: 32h Practical

Course Objectives:

This course aims to provide students with an understanding of logic circuits and computer architecture, including principles of digital logic design, implementation of combinational and sequential circuits, memory systems, and CPU design.

Syllabus:

Boolean algebra. Logic gates. Combinational circuits. Multiplexers. Decoders. Encoders. Latches and flip-flops. Shift registers. Counters. Memory units. Read-only memory (ROM). Random access memory (RAM). Cache memory. Register transfer level (RTL) design. Arithmetic logic unit (ALU). Instruction set architecture (ISA). Control unit design. Microarchitecture. Pipelining. Memory hierarchy. Input/output systems.

Title of the Course: Physical Education

Number of Credits: 1

Type of the Course: General | Practical

Training hours: 8h Theoretical & 16h Practical

Course Objectives:

Physical Education classes are designed to practice and develop skills in activities that will help you maintain fitness throughout your life. During the year, you will set and monitor personal fitness goals that will aid in the development of a personal fitness plan.

I.A.U



Title of the Course: Engineering Statistics & Probabilities

Number of Credits: 3

Type of the Course: Basic | Theoretical

Training hours: 48h Theoretical

Course Objectives:

By learning this course, students will be able to use basic rules of Probability Theory for real modeling of information problems.

Syllabus:

An introduction to theory of sets, samples and their table display together with average, exponent, middle and variance of conversion and composition, probabilities and the relevant theorems, random variables, intermediate and average and variance of distributions, Poisson's two-phases distributions, geometric difference, normal distribution, distribution of several random variable, random sampling and random numbers, sampling from small society, estimation of statistical parameters, assurance intervals, test 2 presumptive test of decision-making, analysis and variance, regression, correlation, nonparametric methods test, fitting straight line on data.

TextBooks:

1. Walpole and Mayers, Probability and Statistics for Engineering and Scientists, 6th edition, Prentice Hall. 1998
2. R.V. Hogg and T. Elliot, Probability and Statistical Inference, 4th edition, Mac Millan, 1993

Title of the Course: Software Engineering

Number of Credits: 3

Type of the Course: Core | Theoretical

Training hours: 48h Theoretical

Course Objectives:

The purpose of this course is to address the engineering points that must be observed in all stages of software production. In this course, students are introduced to the activities and tools needed to produce a software product.

Syllabus:

Importance and objectives of software engineering; planning; stages and design stages controlling; principles of needs analysis; methods of needs analysis; principles of software design; study of several known methods of design such as data structure, data flow, functional, aimed and immediate analysis, outstanding points of programming languages in viewpoint of software engineering, software accuracy and confidentiality, software maintenance, performance of a practical project.

TextBooks:

- 1-Software Engineering, A Practical Approach, Roger S. Pressman, R. S. Pressman, 1987



Title of the Course: Research and presentation methods

Number of Credits: 2

Type of the Course: Specialized | Theoretical

Training hours: 32h Theoretical

Syllabus:

Different types of scientific and technical subjects (letters, reports, pamphlets, manual and etc.), common points in all scientific and technical writings: specifying the objective of writing and its eventual readers, organizing the subjects, abstract of essay together with report, the role of a good introduction, dividing the subjects into parts and chapters, discussion and conclusion, preparing source and reference index, attachments, preparing the pictures and diagrams and tables. Important points in translation of scientific and technical subjects, writing style, marking and its importance, preparing final format of writing by typing machine or computer, foot-article, notes and other lateral subjects, an introduction to research methods, presenting subjects orally, effective use of audio-visual devices.

Title of the Course: Database Systems

Number of Credits: 3

Type of the Course: Core | Theoretical

Training hours: 48h Theoretical

Course Objectives:

Understanding the fundamental concepts of database management systems, relational databases, SQL querying, normalization, database design, and transaction management.

Syllabus:

Introduction to relational database management systems. Entity-relationship modeling. Relational algebra and calculus. SQL data definition and manipulation. Database design principles. Functional dependencies and normalization. Indexing and hashing. Transaction processing and concurrency control. Database recovery and security. Current trends in database technology.



Title of the Course: Principles of Compiler Design

Number of Credits: 3

Type of the Course: Specialized | Theoretical

Training hours: 48h Theoretical

Course Objectives:

Designing and building compilers is one of the basic concepts of computer science. Although there is little variety in the way compilers are made, a wide variety of languages and machines can be used to make interpreters and translators. In this lesson, the subject of building compilers is introduced by describing the main components of a compiler, their tasks and their relationship. After an introductory introduction to the components of a compiler and the types of grammars, the various stages of translation, such as lexical, syntactic and semantic analysis, and code generation and payment, are described. Students who successfully pass this course will have a good insight into the following: Familiarity with compiler components and various techniques for their implementation, understanding the implementation of programming language commands, gaining skills in producing optimal programs and fixing programming errors, familiarity And the use of automated tools in compiler production.

Syllabus:

An introduction to the design and implementation of programming language translators. Theoretical aspects of language design and translation is discussed and practically demonstrated by developing a working compiler.

TextBooks:

A. Aho, M. Lam, R. Sethi, and J. Ullman, Compilers: Principles, Techniques, and Tools (2nd Edition), Prentice Hall, 2006.



Title of the Course: Computer Networks

Number of Credits: 3

Type of the Course: Core | Theoretical

Training hours: 48h Theoretical

Course Objectives:

Basics and Principles of Computer Networks and Data Transition Systems will be introduced in this course. In this course the students will be introduced to different network security algorithms and they can learn about methods of implementation a secure network and a communication plan.

Syllabus:

Structure of networks, network architectures, reference model of ISO, networks of ARPA, SNA, DECNET and general. Network topology, connectivity analysis, delay analysis, design of network with local access. Design of physical layer, fundamentals of theory for data transfer, transfer telephone systems and multiplexing, survey on terminal, transfer errors. Data relation layer, primary protocols for data relation, sliding window protocol, protocols analysis. Primary layer of network, point-point networks, routing algorithms, density. Secondary layer, satellite and radio networks, broadcasting satellite packages, radio packages.

TextBooks:

1. Computer Networking: a top-down approach / James F. Kurose, Keith W. Ross. 6th ed, 2013.
2. Computer networks / Andrew S. Tanenbaum, David J. Wetherall. 5th ed, 2011.

Title of the Course: Sport I

Number of Credits: 1

Type of the Course: General | Practical

Training hours: 32h Practical

Course Objectives:

This course is designed for the prospective coach, physical education and/or recreation professional.



Title of the Course: Computer Workshop

Number of Credits: 1

Type of the Course: Core | Practical

Training hours: 48h Practical

Course Objectives:

The purpose of this course is to provide knowledge and skills necessary for general computer use, which must be passed at the beginning of the course. In this course, the student learns general skills in working with computers, including knowledge of hardware, operating system and common software, web programming, general principles of the network and the Internet and how to use them. The presentation method of this lesson is workshop and scientific, and in each session, concepts are taught to the students in a practical way by the lecturer, and the students learn the presented material through practical exercises. In order to do practical work, it is necessary to install and prepare the software requirements of the taught materials before each session by the workshop manager or the students themselves.

Title of the Course: Operating Systems - Lab

Number of Credits: 1

Type of the Course: Specialized | Practical

Training hours: 32h Practical

Course Objectives:

In this laboratory, which is taken simultaneously with the operating systems course, the aim is to test and scientifically test the concepts in which the course is presented. The first part of this lab is mainly focused on working with the Linux operating system, and in the second part, programming in this operating system and interaction with its core.

Syllabus:

Compile and install Linux. Programming with C++ language and shell in Linux. Using Linux system calls in programs. Checking the behavior of the operating system (proc/path). Create, execute and destroy processes and threads (using the pthread library). Synchronization and communication between processes and threads. Memory management, shared memory and virtual memory. CPU Scheduling. Using Installable File Systems. Disk planning and input and output devices (IO Scheduling). Device Driver design and implementation. Using Linux security mechanisms. Familiarity with real-time and embedded operating systems. Introduction to Windows Research Kernel.



Title of the Course: Programming Language Design

Number of Credits: 3

Type of the Course: Specialized | Theoretical

Training hours: 48h Theoretical

Course Objectives:

This course introduces students to the concepts and components of programming languages and allows them to design and implement a new programming language. Also, understanding the concepts of programming languages helps students to learn new languages more easily. Learning the capabilities of each category of languages along with important examples from each category is also one of the objectives of this course.

Syllabus:

The projects focus either on various aspects of programming languages (for example, simple parsers, translators, symbolic computation, and implementation of abstract data types) or on exercising the particular strengths of a given language. Students work individually or in small groups on several programming projects. Students design, implement, and test their solutions. Each project typically uses a different language, such as: Ada, C++, Java, Smalltalk, Python, LISP, Scheme, Standard ML, Haskell, and Prolog. At least one project language will support object-oriented programming and at least one will be a non-imperative language. Students will: (1) learn the fundamental principles of modern computer programming languages; (2) learn the commonalities and differences among the different languages; (3) learn about a variety of different programming languages and about their relative strengths and weaknesses; and (4) gain experience designing and writing programs in a selected set of languages.

TextBooks:

- 1- J. C. Mitchell, Concepts in Programming Languages, Cambridge University Press, 2002.
- 2- R. W. Sebesta, Concepts of Programming Languages, 11th Edition, Pearson, 2015.
- 3- B. C. Pierce, Types and Programming Languages, MIT Press, 2002.

Title of the Course: Digital Systems Computer Design

Number of Credits: 3

Type of the Course: Specialized | Theoretical

Training hours: 48h Theoretical

Course Objectives:

Students in the Digital Systems Design course will implement and test digital circuits ranging in complexity from basic logic gates to state machines that perform useful functions like calculations, counting, timing, and a host of other applications.



Title of the Course: Computer Networks - Lab

Number of Credits: 1

Type of the Course: Core | Practical

Training hours: 32h Practical

Course Objectives:

The objective of this lab course is to get practical knowledge of working principles of various communication protocols. Analyse structure and formats of TCP/IP layer protocols using network tools such as Wireshark and network simulators.

TextBooks:

1- J. F. Kurose, Keith W. Ross, Computer Networking: A Top-Down Approach, 7th Edition, Pearson, 2016

Title of the Course: Internship

Number of Credits: 1

Type of the Course: Training | Practical

Training hours: 32h Practical

Course Objectives:

Students must practically get involved in industry and market by working totally 240 hours in an appropriate and defined company in either governmental or private sectors as an intern.

I.A.U



Title of the Course: Basics of data mining

Number of Credits: 3

Type of the Course: Specialized | Theoretical

Training hours: 48h Theoretical

Course Objectives:

This course is an introductory course on data mining. It introduces the basic concepts, principles, methods, implementation techniques, and applications of data mining, with a focus on two major data mining functions: (1) pattern discovery and (2) cluster analysis.

Syllabus:

Overview of Data Mining and its applications. Key concepts: Patterns, Models, and Algorithms. Introduction to popular data mining tools. Data Preprocessing: Data cleaning and integration, Handling missing data, Feature scaling and selection. Classification Techniques: Decision Trees, Naive Bayes, Support Vector Machines. Clustering Techniques: K-Means Clustering, Hierarchical Clustering, DBSCAN. Association Rule Mining: Apriori algorithm, Association rule evaluation. Regression Analysis: Linear regression, Non-linear regression models. Ensemble Learning: Bagging and Boosting, Random Forests. Applications in Computer Engineering: System optimization using data mining, Predictive modeling for performance analysis. Integration with Big Data Technologies: Overview of Hadoop and Spark, Data mining in a big data environment.

TextBooks:

- 1- J. Leskovec, A. Rajaraman, and J.D. Ullman. Mining of massive datasets. Cambridge University Press, 2014.
- 2- J. Han, J. Pei, and M. Kamber. Data mining: Concepts and techniques. Elsevier, 2011.

Title of the Course: Implementation of Database system

Number of Credits: 3

Type of the Course: Optional | Theoretical

Training hours: 48h Theoretical

Course Objectives:

Implementing components of relational database systems (DBMS): record storage, indexing structures, query evaluation, joins algorithms, query optimization. Understanding and administering a DBMS: security, concurrency control and crash recovery. Tuning DBMS for performance. Recent advances in data management : text-based information retrieval, web search, cloud computing, column store systems. This class focuses on data management from a database administrator's, or implementer's, perspective.



Title of the Course: Network Security

Number of Credits: 3

Type of the Course: Core | Theoretical

Training hours: 48h Theoretical

Course Objectives:

In this course the students will be introduced to different network security algorithms and they can learn about methods of implementation a secure network and a communication plan.

Syllabus:

Basic concepts and definitions: Security policy and access control models. Hidden channels, information flow control. Optional models (DAC) and mandatory models (MAC). Role-based models (RBAC). System Security: How software is implemented and how it interacts with the system and its weaknesses. Attacks and defensive methods (control hijacking). Secure management of old code in use (sandbox, virtualization, isolation in different layers). Existing methods for secure code development (static analysis, dynamic analysis). Methods of Security Violation and Fuzzing. Web security model: security of web applications (sql, XSS, CSRF). Manage cookies. Concepts of symmetric and asymmetric cryptography. Message authentication codes and hash functions. Security of web information during exchange (Https / SSL). Browser defense mechanisms (SOP, CSP, CORS). Network security: Security threats in network protocols (etc, routing, BGP, DNS, TCP). Network defense tools (etc, IDS, VPN, Firewall). Denial of service attacks and defenses. Trusted Computing and SGX. Mobile security. Security of mobile platforms (iOS, Android). Threats in the field of mobile.

TextBooks:

- 1- Matt Bishop. Computer Security. Addison-Wesley, 2017.
- 2- John Erickson. The Art of Exploitation 2nd Edition, No Starch Press, 2008.
- 3- Robert C. Seacord. Secure Coding in C and C++. 2nd Edition, Pearson Education, 2005.
- 4- A. Sotirov. Bypassing Browser Memory Protections. 2008.



Title of the Course: Basics of Secure Computing

Number of Credits: 3

Type of the Course: Elective | Theoretical

Training hours: 48h Theoretical

Course Objectives:

This course covers fundamental issues and first principles of security and information assurance. The course will look at the security policies, models and mechanisms related to confidentiality, integrity, authentication, identification, and availability issues related to information and information systems.

Title of the Course: Thematic Interpretation of Nahj al-balaghah

Number of Credits: 2

Type of the Course: General | Theoretical

Training hours: 32h Theoretical

Course Objectives:

Getting to know Nahj al-Balagha and understanding its basic teachings with a topical approach.

Title of the Course: Imam Khomeini's Thoughts and Wills

Number of Credits: 1

Type of the Course: Core | Theoretical

Training hours: 16h Theoretical

Course Objectives:

This course aims to introduce students to the philosophical thoughts and advices of Imam Khomeini, emphasizing on their relevance and impact in various aspects of life.

I.A.U



Title of the Course: Holy Quran

Number of Credits: 1

Type of the Course: General | Theoretical

Training hours: 16h Theoretical

Course Objectives:

Teaching Quran Reading

Title of the Course: Signals and Systems

Number of Credits: 3

Type of the Course: Core | Theoretical

Training hours: 48h Theoretical

Course Objectives:

Primary definitions of systems and signal, various kinds of systems, introduction to modeling of various physical systems, analysis of linear and time independent (continuous and discrete) systems, impulse response, convolution integral, Fourier analysis, energy density, spectrum and power sampling theorem, system analysis by Laplace transform, signal flow graphs, system analysis in state space (continuous and discrete), Z transform, discrete systems analysis by the Z transform. The aim of this course is to introduce students to the modeling, description and analysis of signals and systems in time and frequency domains from both theoretical and practical perspectives. MATLAB exercises are presented in addition to theoretical exercises for practical mastery of concepts.

TextBooks:

1- A.V. Oppenheim, A.S. Willsky, S.H. Nawab, Signals and Systems, 2nd Edition, Prentice-Hall, 1997.

I.A.U



Title of the Course: Basic systems security

Number of Credits: 3

Type of the Course: Optional | Theoretical

Training hours: 48h Theoretical

Course Objectives:

This course covers fundamental issues and first principles of security and information assurance. The course will look at the security policies, models and mechanisms related to confidentiality, integrity, authentication, identification, and availability issues related to information and information systems.

Title of the Course: Internet Engineering

Number of Credits: 3

Type of the Course: Core | Theoretical

Training hours: 48h Theoretical

Course Objectives:

This course provides an introduction to fundamental concepts of the Internet and web technologies including their architecture, protocols and applications. The main topics of the course include: Introduction to Internet history and Internet services, networking basics, overview of TCP/IP (addressing, routing and transport protocols), network programming, web programming, web services, web servers and Internet security.

TextBooks:

Internet & World Wide Web How to Program, 4th edition, Harvey M. Deitel and Paul J. Deitel, , Prentice Hall , 2008.

Title of the Course: Microprocessor Lab - 1

Number of Credits: 1

Type of the Course: Core | Practical

Training hours: 32h Practical

Course Objectives:

This course introduces the assembly language programming of 8086 and 8051 microcontroller. it gives a practical training of interfacing the peripheral devices with the 8086 microprocessor. The course objective is to introduce the basic concepts of microprocessor and to develop in students the assembly language programming skills and real time applications of Microprocessor as well as microcontroller.



Title of the Course: Information Security Engineering and Management

Number of Credits: 3

Type of the Course: Specialized | Theoretical

Training hours: 48h Theoretical

Course Objectives:

The goal of this course is to familiarize students with how to manage and engineer diverse architectures, processes, and actors with different motivations to achieve security. The first part of the course addresses personal security frameworks and examines related management and engineering processes. The second part examines the issue of security in companies in terms of existing frameworks, risks, and how to manage users, infrastructures, and systems. At the end of the course, policymaking in national and international dimensions and important actors are reviewed.

Syllabus:

Information security frameworks and policies. Personal security frameworks. Trust / Do you own your device? TPM and trusted computing. Open devices, open firmware, and open architectures. Protection of personal devices. Secure update of storage / secure data deletion. Disk-level or file system-level encryption. Data and identity protection. Identifiers in smartphones (DMV identifier in Android). Two-factor authentication (2FA) and time-based one-time passwords (TOTP). Secret sharing as a service. Human factors and usable security. Enterprise-level security frameworks. Corporate security frameworks. Enterprise security architectures (SBSA, DoDAF, TOGAF). Case studies of security frameworks in specific domains. Security in financial services and the healthcare sector. Risk analysis (NIST Cybersecurity Framework). Security management in enterprises. Devices versus identities. BYOD and device management. Identity and Access Management (IAM). Software supply chain security. RFID in various applications. Open-source software and digital rights management (DRM). National and international-level security frameworks. International dimensions and stakeholders. Data protection. GDPR. Policy-making and legal evolution. Information security laws: such as HIPAA, GLBA, FISMA. Compliance verification.

TextBooks:

- 1- Anderson, R. (2020). Security engineering: a guide to building dependable distributed systems. John Wiley & Sons.
- 2- Siboni, G and Sivan-Sevilla, I (2019). Regulation in Cyberspace. Institute for National Security Studies.
- 3- Whitman, Michael E., and Herbert J. Mattord. (2013). Management of information security. Nelson Education.
- 4- Schneier, B. (2015). Data and Goliath: The hidden battles to collect your data and control your world. WW Norton & Company.



Title of the Course: Islamic Revolution of Iran

Number of Credits: 2

Type of the Course: General | Theoretical

Training hours: 32h Theoretical

Course Objectives:

Theoretical acquaintance with the causes and factors of the emergence of the Islamic Revolution and the study of the cultural, social and political developments of the Islamic Revolution and the issues after it.

I.A.U



Title of the Course: History of Imamate

Number of Credits: 2

Type of the Course: General | Theoretical

Training hours: 32h Theoretical

Course Objectives:

Familiarity with the issue of Imamate, its place in Islam and the manners and traditions of the leaders in different time requirements.

Syllabus:

Will and succession in the history of previous prophets. Terminology of Imamate, Wilayat, Ahl al-Bayt, Sahaba, Tabi'in, etc. The death of the Prophet and the chain of succession. The story of Saqifa, its contexts and consequences. Life periods of Imam Ali (a.s.) (together with the Prophet, the Caliphate era, the reign period). The biography of Imam Ali (a.s.) and the principles of Alawite government. Imam Hasan (a.s.) during the period of the Prophet and Imam Ali (a.s.). Peace of Imam Hassan (AS), motivations and consequences. Imam Hussein (a.s.) during the period of the Prophet and Imam Ali (a.s.). Ashura uprising, its contexts and effects. Personality and actions of Imam Sajjad (AS). Political, social and cultural characteristics of Imam Baqir (a.s.) and Imam Sadiq (a.s.). The scientific and spiritual personality of Imam Baqir (a.s.) and Imam Sadiq (a.s.) from the point of view of Shiites and Sunnis. The role of Imam Baqir (a.s.) and Imam Sadiq (a.s.) in strengthening and flourishing Islamic and Shiite culture and education. The political and social situation during the time of Imam Kazem (a.s.). The character of Imam Kazem (a.s.) and his influence in the Islamic and Shiite society. The political, social and cultural situation during the era of Imam Reza (a.s.). The personality of Imam Reza (a.s.) and his cultural and political influences in the Islamic society. The issue of Imam Reza's (a.s.) guardianship, motivations, causes and consequences. Social, cultural and political conditions of the era of Imams. The personality of imams and their influence in Islamic and Shiite society. Creating a network of scientific, social and political communication between imams and the Shia community (lawyers, narrators and jurists). Absence contexts and social and political conditions of that period. Imam Mahdi (AS) in Sunni and Shia sources. Shi'ism in the era of Ghaib Soghari and Nawab Arbaah. The philosophy of expectation and conditions of emergence.



Title of the Course: Software Project

Number of Credits: 3

Type of the Course: Specialized | Practical

Training hours: 96h Practical

Course Objectives:

student choose a professor of computer science department and he/she defines a subject such as implementing a program on special scientific usage or a subject in computer software to research about that, and student must provide an essay and submit their project result and present it.

Title of the Course: Advanced Programming

Number of Credits: 3

Type of the Course: Core | Theoretical

Training hours: 48h Theoretical

Course Objectives:

The purpose of this course is to study the principles and methods necessary to produce a computer program with good quality specifications. In this regard, after covering the top-down design method to solve the problem, students will be introduced to the concepts and techniques of object-oriented programming as a tool for managing complexity in medium and large-scale applications. Program performance, testing, and debugging are focused throughout the course. An object-oriented programming language (such as Java) is used to teach these principles and methods, and students are introduced to the concepts and techniques of object-oriented programming and module-based object design, advanced concepts such as graphical user interface design. Multidisciplinary programming (parallel) and distributed programs on the network are also covered in this course.

Syllabus:

In depth Java programming, Introduction of UI design QT. Complementary issues of Java programming, Memory management, in depth understanding of Java codes, Coding relation with operating system, file management, IO streams, clear implementing basic data structures like link lists, Generic programming, Implementation of inheritance and its related issues in Java. Multithreading essentials, exception handling, object oriented programming principles, Debugging and testing of programs, Function calling conventions, Dynamic memory coding.

TextBooks:

Java : how to program / P.J. Deitel, H.M. Deitel. -- 9th ed, 2013.